Super Secret Gossip Phone — Feather ESP32-S3 carrier board

Feather plugs into J1/J2. Breakout boards (microSD, I2S mic, I2S amp) plug into sockets J3–J5.

Pins match mp3_recorder.ino: SD CS=D10, I2S BCLK=D9, WS=D6, mic data=D12, amp data=D5, buttons A0/A1/A2.

J1
+3V3 1 3V3
2 3V3
3 AREF
GND 4 GND
BTN_REC 5 A0
BTN_PLAY 6 A1
BTN_NEXT 7 A2
ENC_A 8 A3
ENC_B 9 A4
ENC_SW 10 A5
SD_SCK 11 SCK
SD_MOSI 12 MOSI
SD_MISO 13 MISO
14 RX
15 TX
16 NC
Feather 16-pin socket

J2
1 BAT
2 EN
+5V 3 USB
MODE 4 D13
I2S_MIC 5 D12
LED_DRV 6 D11
SD_CS 7 D10
I2S_BCLK 8 D9
I2S_WS 9 D6
I2S_SPK 10 D5
I2C_SCL 11 SCL
I2C_SDA 12 SDA
Feather 12-pin socket

J3
1 5V
+3V3 2 3V
GND 3 GND
SD_SCK 4 CLK
SD_MISO 5 DO
SD_MOSI 6 DI
SD_CS 7 CS
8 CD
microSD breakout (Adafruit 254 pinout)

J4
+3V3 1 3V
GND 2 GND
I2S_BCLK 3 BCLK
I2S_MIC 4 DOUT
I2S_WS 5 LRCL
GND 6 SEL
I2S mic breakout (SPH0645)

J5
AMP_VIN 1 VIN
GND 2 GND
3 SD
4 GAIN
I2S_SPK 5 DIN
I2S_BCLK 6 BCLK
I2S_WS 7 LRC
I2S amp breakout (MAX98357A)

J7
GND 1 GND
+5V 2 VCC
I2C_SDA 3 SDA
I2C_SCL 4 SCL
20x4 I2C LCD (backpack)

J8
LISTEN 1 GND
COM 2 MODE
RECORD 3 +3V3
Mode toggle (on/on)

JP1
5V (USB) 1 +5V
to amp VIN 2 AMP_VIN
3V3 3 +3V3
Amp power select

Shunt 1-2: amp on USB 5 V (louder). Shunt 2-3: amp on 3.3 V.

CAMP_VIN
+
GND
10uF

J6
LED + 1 LED_A
LED - 2 GND
External recording LED

SW1
BTN_REC GND
Record / Stop

SW2
BTN_PLAY GND
Play / Stop

R1
LED_DRV
SW3
BTN_NEXT GND
LED_A
330R
Next file

ENC1
ENC_A A S1 ENC_SW
GND C
ENC_B B S2 GND
EC11 dial

H1 M2.5
H2 M2.5
H3 M2.5
H4 M2.5

GND
#PWR_FLAG
+3V3
#PWR_FLAG
+3V3
+5V
#PWR_FLAG
+5V

gregariouspanda
Sheet: /
File: supersecretgossipphone.kicad_sch
Title: Super Secret Gossip Phone carrier
Size: A3 | Date: | Rev: A
KiCad E.D.A. 10.0.7 | Id: 1/1